

Incomplete Exam for the Second Midterm

$$\gamma_1 \cdot f \wedge \gamma_2 \cdot f \gamma_1 \cdot \gamma_2, \quad \gamma_1 \gamma_2 \supseteq \rho$$

1. Which conclusion is correct about a series whose n -th partial sum is

(4 Points)

$$s_n = \frac{7-11n}{3n+11}$$

- ☐ It is convergent with sum $-\frac{11}{3}$. ✓
- ☐ It is convergent with sum $\frac{7}{3}$.
- ☐ It is convergent with sum 11.
- ☐ It is divergent.

2. (4 Points)

If the n -th partial sum of the series $\sum_{n=1}^{\infty} a_n$ is $s_n = \frac{2n-1}{n+2}$, find a_n .

- ☐ $a_1 = \frac{1}{3}$, $a_n = \frac{5}{(n+1)(n+2)}$ ($n \geq 2$) ✓
- ☐ $a_1 = \frac{1}{3}$, $a_n = \frac{5}{(n+2)(n+3)}$ ($n \geq 2$)
- ☐ $a_1 = \frac{3}{2}$, $a_n = \frac{5}{(n+1)(n+2)}$ ($n \geq 2$)
- ☐ $a_1 = \frac{3}{2}$, $a_n = \frac{5}{(n+2)(n+3)}$ ($n \geq 2$)

3. Which of the following is correct for the geometric series

(4 Points)

$$1 + 0.4 + 0.16 + 0.064 + \dots$$

- ☐ It is convergent with sum $\frac{5}{3}$. ✓
- ☐ It is convergent with sum $-\frac{3}{5}$.
- ☐ It is convergent with sum $-\frac{1}{3}$.
- ☐ It is divergent.

4. Which of the following is true for the series

(4 Points)

$$\sum_{n=1}^{\infty} \frac{e^n}{3^{n-1}}$$

- ☐ It is convergent with sum $\frac{3e}{3-e}$. ✓
- ☐ It is convergent with sum $\frac{e}{3-e}$.
- ☐ It is convergent with sum $\frac{3e}{1-e}$.
- ☐ It is divergent.

5. Which of the following is true for the series

(4 Points)

$$\sum_{n=3}^{\infty} \frac{2n^2-1}{(n-2)(n-1)}$$

- ☐ It is divergent. ✓
- ☐ It is convergent with sum $\frac{5}{6}$.
- ☐ It is convergent with sum $\frac{3}{2}$.
- ☐ It is convergent with sum $\frac{1}{6}$.

6. Which of the following arguments is correct for the series

(4 Points)

$$\sum_{n=1}^{\infty} \cos\left(\frac{5}{n}\right)$$

- ☐ It is divergent since $\lim_{n \rightarrow \infty} \cos\left(\frac{5}{n}\right) \neq 0$. ✓
- ☐ It is divergent since $\sum_{n=1}^{\infty} \frac{5}{n}$ is divergent and $\cos$ is a continuous function.
- ☐ It is convergent by the Alternating Series Test.
- ☐ It is convergent by the Integral Test.

7. Which of the following is correct for the series

(4 Points)

$$\sum_{n=1}^{\infty} \frac{1}{(n-2)^2+1}$$

- ☐ The Integral Test applies and concludes that the series is divergent.
- ☐ The Integral Test applies and concludes that the series is convergent. ✓
- ☐ $f(x) = \frac{1}{(x-2)^2+1}$ is not decreasing on $[2, \infty)$, so The Integral Test does not apply.
- ☐ $f(x) = \frac{1}{(x-2)^2+1}$ is not positive on $[2, \infty)$, so The Integral Test does not apply.

8. (4 Points)

The series $\sum_{n=1}^{\infty} a_n$ converges if and only if

- ☐ The sequence of partial sums $\{S_n\}$ converges. ✓
- ☐ $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| > 1$
- ☐ $\lim_{n \rightarrow \infty} a_n = 0$
- ☐ The sequence of partial sums $\{S_n\}$ is bounded.
- ☐ The given series is alternating.

9. Find all positive values of c for which the following series converges.

(4 Points)

$$\sum_{n=1}^{\infty} e^{c \ln n}$$

- ☐ $0 < c < \frac{1}{e}$. ✓
- ☐ $c = e$
- ☐ $0 < c < e$
- ☐ $c > \frac{1}{e}$

10. Which of the following is correct for the series

(4 Points)

$$\sum_{n=0}^{\infty} \frac{n^2+1}{n^4+1}$$

- ☐ It is convergent by the Limit Comparison Test with the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$. ✓
- ☐ It is convergent by the Comparison Test with the p -series with $p = \frac{1}{2}$.
- ☐ It is divergent by the Comparison Test with the p -series with $p = 1$.
- ☐ It is divergent by the Test for Divergence.

11. Which of the following is correct for the series

(4 Points)

$$\sum_{n=1}^{\infty} \frac{e^n}{n^{1+\frac{1}{2}}}$$

- ☐ It is divergent using the Limit Comparison Test with the series $\sum_{n=1}^{\infty} \frac{1}{n}$. ✓
- ☐ It is convergent using the (Basic) Comparison Test with the series $\sum_{n=1}^{\infty} \frac{e}{n^2}$.
- ☐ It is divergent by the Test for Divergence.
- ☐ It is convergent using the Limit Comparison Test with the series $\sum_{n=1}^{\infty} \frac{1}{n^{1.1}}$.

12. Which of the following statements is false?

(4 Points)

- ☐ If $a_n > 0$ and $\lim_{n \rightarrow \infty} n^2 a_n = 2$, then $\sum_{n=1}^{\infty} a_n$ is divergent. ✓
- ☐ If $a_n > 0$ and $\lim_{n \rightarrow \infty} n a_n = 2$, then $\sum_{n=1}^{\infty} a_n$ is divergent.
- ☐ If $a_n > 0$ and $\sum_{n=1}^{\infty} a_n$ is convergent, then $\sum_{n=1}^{\infty} \sin(a_n)$ is convergent.
- ☐ If $a_n > 0$ and $\sum_{n=1}^{\infty} a_n$ is convergent, then $\sum_{n=1}^{\infty} \cos(a_n)$ is divergent.

13. Find all values of p for which the following series converges.

(4 Points)

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{(\ln k)^p}$$

- ☐ $p > 0$. ✓
- ☐ $p < 0$
- ☐ $p > -1$
- ☐ $p > 1$

14. (4 Points)

Find ALL values of x for which the series $\sum_{n=1}^{\infty} \frac{3+x^n}{7^n}$ is convergent

- ☐ $-7 < x < 7$. ✓
- ☐ It is not possible because the series is Divergent.
- ☐ $-3 < x < 7$
- ☐ $-3 < x < 3$
- ☐ $|x| < 1$

15. Which of the following is correct for the series

(4 Points)

$$\sum_{n=1}^{\infty} \frac{n^n}{10^{9n+1}}$$

- ☐ It is divergent by the Root Test. ✓
- ☐ It is divergent by comparison with a p -series.
- ☐ It is convergent by the Ratio Test.
- ☐ It is convergent by comparison with a geometric series.

16. Which one of the following series is absolutely convergent?

(4 Points)

- ☐ $\sum_{k=1}^{\infty} \frac{(-1)^k \arctan k}{k^4}$. ✓
- ☐ $\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{\sqrt[4]{k}}$
- ☐ $\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k}$
- ☐ $\sum_{k=1}^{\infty} \frac{\cos k\pi}{k^{-3}}$

17. Determine which of the series below is conditionally convergent.

(4 Points)

- ☐ $\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{\sqrt[4]{k}}$. ✓
- ☐ $\sum_{k=1}^{\infty} \frac{(-1)^k \arctan k}{k^4}$
- ☐ $\sum_{k=1}^{\infty} \frac{(-1)^k k^3}{3^k}$
- ☐ $\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k^{-2}}$

18. Which of the following is correct for the series

(4 Points)

$$\sum_{n=1}^{\infty} (-1)^n \tan\left(\frac{1}{n}\right)$$

- ☐ It is conditionally convergent. ✓
- ☐ It is absolutely convergent.
- ☐ It is divergent.
- ☐ None of these.

19. (4 Points)

If the n th partial sum of a series $\sum_{n=1}^{\infty} a_n$ is $S_n = 5 - \frac{3}{n}$, then $a_3 =$

- ☐ $\frac{1}{2}$. ✓
- ☐ 2
- ☐ $\frac{5}{3}$
- ☐ -1
- ☐ 4

20. Find the radius of convergence of the power series.

(4 Points)

$$\sum_{n=1}^{\infty} \frac{n! x^n}{9^n}$$

- ☐ 9. ✓
- ☐ ∞
- ☐ $\frac{1}{2}$
- ☐ 3
- ☐ $\frac{1}{9}$

21. Find the interval of convergence of the power series

(4 Points)

$$\sum_{n=1}^{\infty} \left(\frac{x}{3}\right)^n$$

- ☐ $(-3, 3)$. ✓
- ☐ $(-3, 3]$
- ☐ $(-\infty, \infty)$
- ☐ $(-\frac{1}{3}, \frac{1}{3})$
- ☐ $\{0\}$

22. Use the power series representation centered at 0 of $f(x)$ to find the sum s .

(4 Points)

$$f(x) = \frac{1}{(1-x)^2}, \quad s = \sum_{n=1}^{\infty} \frac{n}{3^{n-1}}.$$

- ☐ $s = \frac{9}{4}$. ✓
- ☐ $s = \frac{1}{4}$
- ☐ $s = \frac{3}{2}$
- ☐ $s = \frac{1}{2}$

23. Use the power series representation centered at 0 of $\arctan(x)$ to find the sum of the series

(4 Points)

$$\sum_{n=0}^{\infty} (-1)^n \frac{2^{2n+1}}{(2n+1)7^{2n}}$$

- ☐ $7 \arctan\left(\frac{2}{7}\right)$. ✓
- ☐ $\frac{7}{2} \arctan\left(\frac{7}{2}\right)$
- ☐ $2 \arctan\left(\frac{2}{7}\right)$
- ☐ $\arctan\left(\frac{7}{5}\right)$

24. Which of the following pairs is a power series representation and its radius of convergence for the function

(4 Points)

$$f(x) = \frac{1}{9x^2+1}$$

- ☐ $\sum_{k=0}^{\infty} (-1)^k 3^{2k} x^{2k}, \quad R = \frac{1}{3}$. ✓
- ☐ $\sum_{k=0}^{\infty} (-1)^k 3^k x^{2k+1}, \quad R = \frac{1}{3}$
- ☐ $\sum_{k=0}^{\infty} (-1)^k 2^{2k} x^{k+1}, \quad R = 4$
- ☐ $\sum_{k=0}^{\infty} (-1)^k 2^k x^{k+1}, \quad R = 2$

25. Which pair is a power series representation with its radius of convergence for the indefinite integral

(4 Points)

$$\int \ln(1+x^4) dx$$

- ☐ $C + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^{4n+1}}{n(4n+1)}, \quad R = 1$. ✓
- ☐ $C + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^{4n+1}}{4^{n+1}}, \quad R = 4$
- ☐ $C + \sum_{n=1}^{\infty} \frac{x^{n+2}}{n(n+4)2^n}, \quad R = 1$
- ☐ $C + \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^{4n}}{(n+1)^n}, \quad R = 1$